



Nutrition for Competitive Swimmers

Nutrition in Action
Steps to a Healthy You

*Reliable Nutrition Advice
From Atlantic Canada's
Leading Dietitian*

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ABOUT ANGELA!

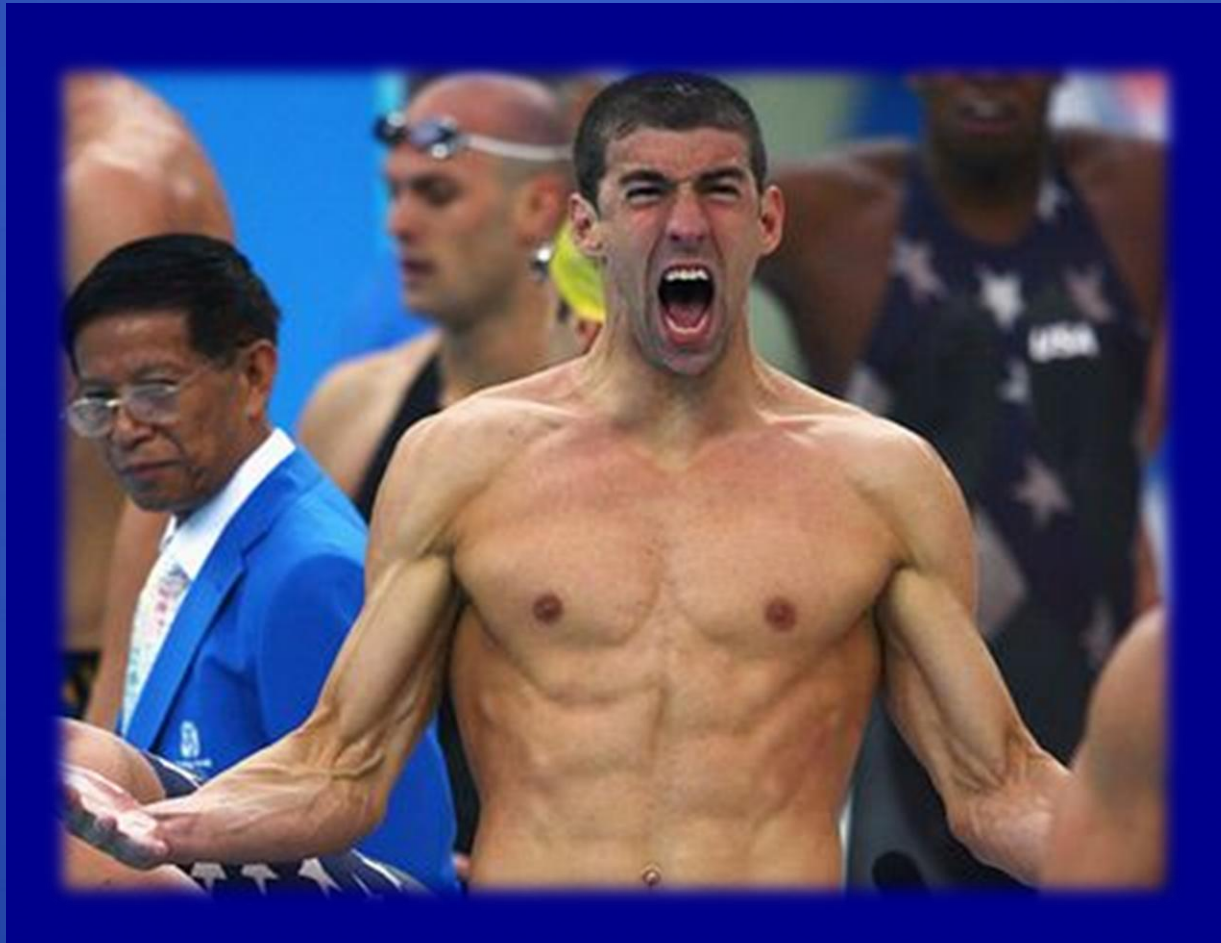
- ❖ Professional (Sports) Dietitian
- ❖ Bachelor of Science in Human Ecology (BScHE) Nutrition
- ❖ Masters degree in Adult Education
- ❖ International Olympic Committee's (IOC) Graduate
- ❖ Level 2 Anthropometrist



Nutrition in Action
Steps to a Healthy You

*Reliable Nutrition Advice
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“A proper diet cannot make an average athlete elite
but a poor diet can make an elite athlete average” -
Costill 1983



Outline



- Energy requirements for swimmers:
CHO/PRO/Fat requirements
- Nutrition before, during & after training/
competition
- Hydration

Proper Nutrition Can:

Competitive athletes may train 300-600 times each year, but most will eat 1200-1600 times in the same period. Therefore, well-chosen eating practices have much to offer the athlete:

- Fuel to train and perform at a competitive level
- Optimum gains from the training program
- Enhanced recovery between workouts and between events
- Achievement and maintenance of an ideal body weight and physique
- Increase immunity
- Reduced risk of injury, overtraining fatigue and illness
- Sustain concentration and mental skills over the day
- Confidence in being well-prepared to face competition
- Achieve consistent high-level competition performances
- Enjoyment of food and social eating occasions at home and during travel



Nutrition Challenges

- School & Other Activities
- Budget?
- Shopping/Cooking skills?
- Travel/On the Road Eating
- Training Schedules:
 - Daily – often several times/day
- Competition Schedules:
 - 3 day meets/invitationals
 - Multiple events per day



Key Nutrient Needs

- Carbohydrates – fuel for your gas tank
 - ▶ This nutrient is your main source of energy
- Protein – the mechanic
 - ▶ Used for recovery, building/rebuilding muscles
- Fluids – oil to keep your engine from seizing
 - ▶ Fluids keep nutrients moving through your body

Nutrition in Action,
Professional Education Services

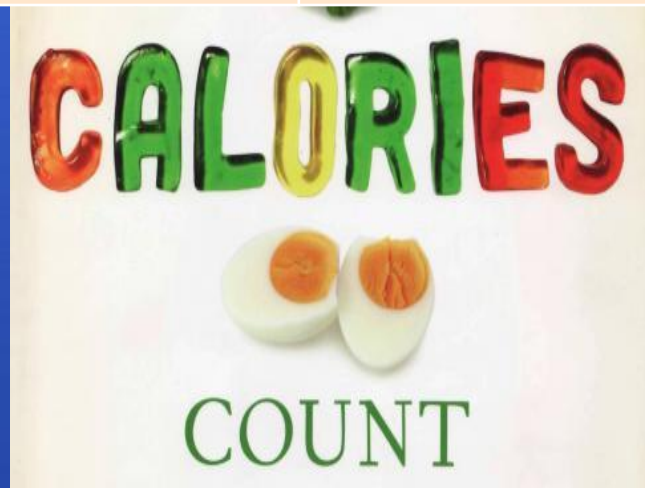


How Much Energy?

- **DID YOU KNOW?.....**Studies have shown that male swimmers in high volume training **SHOULD** consume about 4,000-5,000kcal daily (48kcal/kg).
 - Some studies have shown that female swimmers training were not consuming enough food to meet their energy needs.
 - Training camp > energy needs
 - Taper < energy needs
- period)
- 45-52kcal/Kg/day

Estimated Energy Needs for Swimmers

9-13 yrs Sedentary	9-13 yrs Very Active	14-18 Sedentary	14-18 Very Active
1400 kcal/day	2500+ Kcal/day	1700 kcal/day	3000+ kcal/day



Canada's Food Guide

Recommended Number of Food Guide Servings per Day

	Children				Youth				Adults			
	2-3		4-6		7-13		14-18		19-50		51+	
	Age in Years	Sex	Girls and Boys	Girls and Boys	Females	Males	Females	Males	Females	Males	Females	Males
Vegetables and Fruit	4	5	6	7	8	7-8	8-10	7	7			
Grain Products	3	4	6	6	7	6-7	8	6	7			
Milk and Alternatives	2	2	3-4	3-4	3-4	2	2	3	3			
Meat and Alternatives	1	1	1-2	2	3	2	3	2	3			

The chart above shows how many Food Guide Servings you need from each of the four food groups every day.

Having the amount and type of food recommended and following the tips in Canada's Food Guide will help:

- Meet your needs for vitamins, minerals, and other nutrients.
- Reduce your risk of obesity, type 2 diabetes, heart disease, certain types of cancer, and osteoporosis.
- Contribute to your overall health and vitality.

What is One Food Guide Serving?
Look at the examples below.



Oils and Fats

- Include a small amount – 30 to 45 mL (2 to 3 Tbsp) – of unsaturated fat each day. This includes oil used for cooking, salad dressings, margarine and mayonnaise.
- Use vegetable oils such as canola, olive and soybean.
- Choose soft margerines that are low in saturated and trans fats.
- Limit butter, hard margarine, lard and shortening.

Make each Food Guide Serving count...

wherever you are – at home, at school at work or when eating out!

- Eat at least one dark green and one orange vegetable each day.
- Go for dark green vegetables such as broccoli, Brussels sprouts and spinach.
- Go for orange vegetables such as carrots, sweet potatoes and sweet squash.
- Choose vegetables and fruit prepared with little or no added fat, sugar or salt.
- Eat vegetables, fruit, baked or not, frozen or fresh.
- Have vegetables and fruit more often than juice.

- Make at least half of your grain products whole grain each day.

- Eat a variety of whole grains such as barley, brown rice, oats, quinoa and wild rice.
- Enjoy whole grain breads, oatmeal or whole wheat pasta.

- Choose grain products that are lower in fat, sugar or salt.

- Compare the Nutrition Facts to be on label to make better choices.
- Enjoy the full taste of grain products. When adding sources of spirals, use small amounts.

- Drink skim, 1%, or 2% milk each day.

- Have 500 mL (2 cups) of milk and enjoy fat alternative when it.
- Avoid full-fat beverages if you do not drink milk.

- Select low-fat milk alternatives.

- Compare the Nutrition Facts to be on label to make better choices.

- Have meat alternatives such as beans, lentils and tofu often.

- Eat at least two Food Guide Servings of fish each week.*

- Choose fish such as cod, halibut, mackerel, salmon, sardines and trout.

- Select lean meat and alternatives prepared with little or no added fat or salt.

- Avoid visible fat like marinate, brown or dark sauce packets.

- Use cooking methods such as broiling, baking or poaching that require little or no added fat.

- Choose lean meats, sausages or prepackaged meats, choose those lowest in salt (sodium) and fat.



* Health Canada provides advice for limiting exposure to mercury from certain types of fish. Refer to www.hc-sc.gc.ca/nutrition/food/fish/index-eng.php for the latest information.

A Swimmers Food Guide

	Taper/ Light training	Moderate Training/ Comps	Heavy aerobic training
Grains/Starches	6-7	8-12	12-15
Vegetable Fruit	7-8	8-10+	12-15
Milk & Alternatives	2	2-4+	3-4+
Meat and Alternatives	2-3	2-3+	2-6
Fats/oils	2-3	2-3	2-3
Extras	1-2	2-3	3-5

Carbohydrates Needs

Recommended: 3-8 g/kg BW/Day: training dependant

•HOW TO FIGURE THIS OUT?

A 120 lbs. swimmer = 55kg

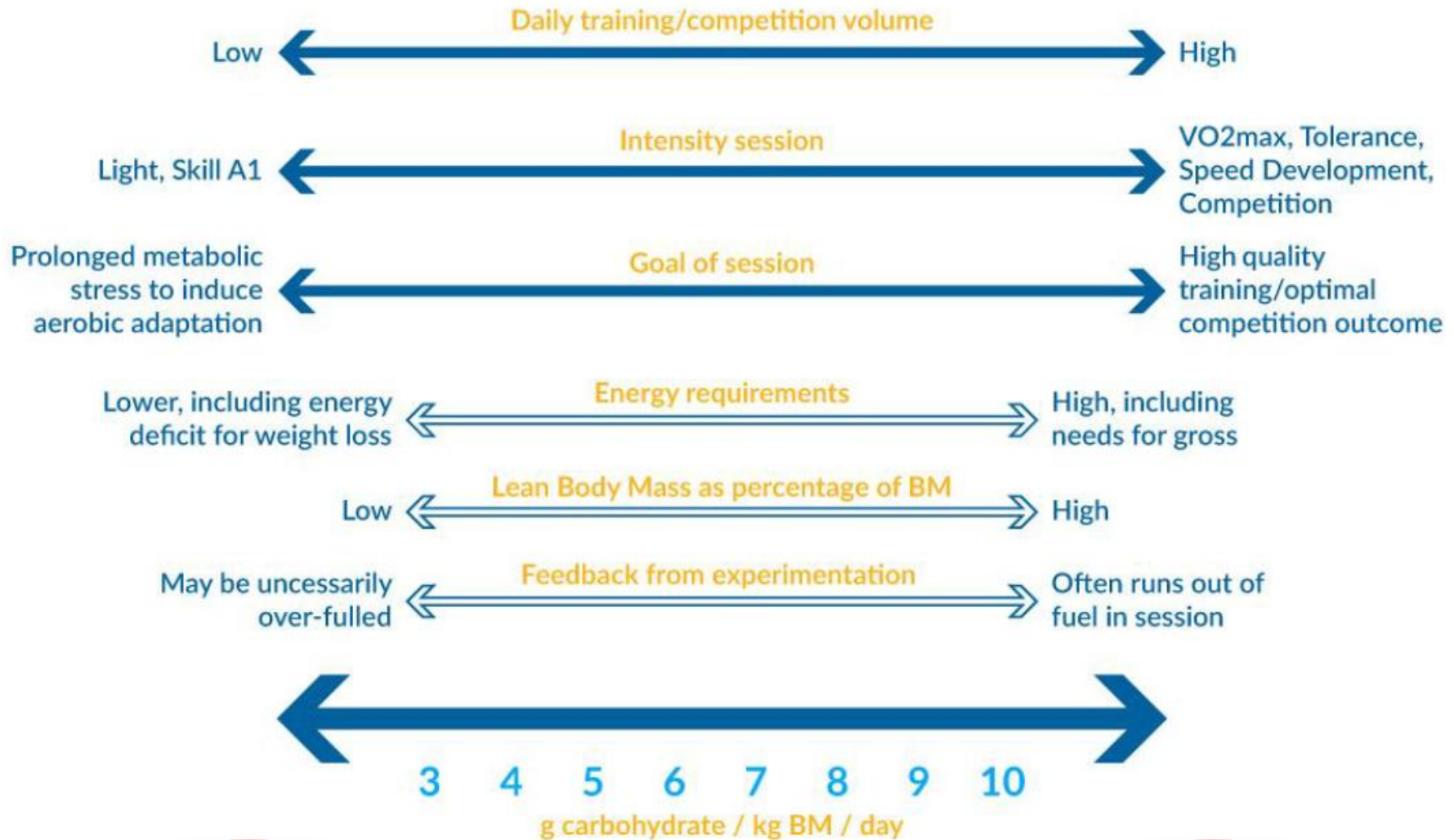
$$55 \times 6 \text{ g/kg} = 330 \text{ g}$$

$$55\text{kg} \times 8 \text{ g/kg} = 440\text{g}$$

What end of the range
would you choose
during taper?

120 lbs. swimmer's CHO range : 330g- 440g

Considerations in setting daily carbohydrate intake targets for aquatic athletes



Muscle glycogen less limiting for completion of session
Less need for carbohydrate intake over total day or around sessions
Some sessions may be deliberately done with low carbohydrate availability

High muscle glycogen requirement for completion of training or optimal competition performance

Promote opportunities for carbohydrate intake in total day and around sessions

Examples of High Carbohydrate Food (30g)

- ✓ 2 med pcs of fruit or 1 large banana
- ✓ 1.5 c cooked oatmeal with milk
- ✓ 1 c yogurt
- ✓ 250ml fruit smoothie or liquid meal supplement (i.e BOOST)
- ✓ 2 slices of bread
- ✓ 1 c baked beans
- 1 c cooked rice



Protein Needs

20-30 g spread out
4-5 x day!!

1.2-1.4 g/kg/BW/day—training
1.4-1.8 g/day- heavy strength training

HOW TO FIGURE THIS OUT?

A 120lbs. Swimmer = 55kg:

55kg swimmer X 1.2g = 66

55kg swimmer X 1.4g = 77g

Recommended intake = 66-77g/kg



Examples of Protein (~10g)

- ✓ 2 small eggs
- ✓ 30g cheese
- ✓ 1 c milk
- ✓ 40g cooked chicken
- ✓ 50g fish/seafood
- ✓ 2 c whole grain cereal
- ✓ 2 c cooked pasta
- ✓ $\frac{3}{4}$ c lentils/beans
- ✓ 60g nuts/seeds



Tapered / Light Training

Typical daily living activity + 60 minutes of daily moderate activity



PERFORMANCE PLATE

To drink
Water, herbal tea, tea, coffee



Fruits
(1 serving)



Milk & Alternatives
(1/2 to 1 serving)



Flavours
Herbs, spices, mustard,
lemon juice



Fats & Oils
(1/2 to 1 serving)



Extras
Plan extras if needed



Moderate training

Typical daily activities
+ 60 to 120 minutes of
moderate to vigorous
activity.



PERFORMANCE PLATE

To drink
Water, herbal tea, tea, coffee



Fruits
(1 to 2 servings)



Milk & Alternatives
(1/2 to 1 serving)



Flavours
Herbs, spices, mustard
lemon juice



Fats & Oils
(1 serving)



Extras
Plan extras if needed
(1 serving)



Created By:



Heavy training

Typical daily activities
+ at least 120 minutes
of vigorous activity



PERFORMANCE PLATE

To drink
Water, herbal tea, tea, coffee



Fruits
(2 to 3 servings)



Milk & Alternatives
(1 to 2 servings)



Flavours
Herbs, spices, mustard
lemon juice



Fats & Oils
(1 to 2 servings)



Extras
Plan extras if needed
(1 to 2 servings)



What to Eat Before

Focus on quality carbohydrates, lean easily digestible proteins and fluid

Meal : 3-4 hours before

200-300 grams of Carbohydrate, some Protein

Snack (solid or liquid): 1-2 hours before

Moderate-High in Carbohydrates

Familiar foods (PRACTICE!)

Food Examples Before Exercise

Time before event	Food choices
3-4 hours	1.5 c pasta with 1 cup tomato sauce 2 pancakes with 2 tbsp syrup and 1 cup chocolate milk
2-3 hours	2 English muffin with 2 tbsp jam 1 cup oatmeal with 1 c fruit juice Cereal bar and a banana
1-2 hours	250 ml Smoothie or Boost
15 minutes	Sport gel or sip on a sport drink

Food/Fluid During Long Training/Between Meets

Focus on easily digested high carbohydrate /mod. protein foods

(min 20-30 g Carbs/hour)

- ✓ Sports drinks/diluted juices
- ✓ Sports gels (w caffeine?)
- ✓ High Carb /moderate protein sports/cereal/granola bars
 - ✓ Cliff/Elevate Me/PowerBars/Chews
- ✓ Fresh or dried fruit of choice
- ✓ Dried Fruit/ Leathers/Squishies



Recovery Nutrition

The 4 R's of recovery

- Refuel
- Repair tissue and muscle damage
- Replace fluid
- Rest

Did you know that in a typical hard two-hour workout, you can use up all your stored carbohydrate energy (muscle and liver glycogen), sweat away over two litres of water (along with approximately 1600 mg of sodium), and break down a variety of different body cells including muscle and red blood cells?



Recovery Nutrition

Phase 1: Immediately (within 20 min)

Phase 2: Within 1-2 hours

Refuel: ~1.2-1.5 g/kg carbohydrates (60-75g)

Repair: 0.3-0.4g/kg g protein. (15-25g)
TYPES? Whey/Leucine

Rehydrate: 1.2-1.5L of fluid for every 1 kg lost

3/4:1 ratio Carbs to Protein!



Recovery Food Choices

Targets: 70-80 g CHO and 15-20g PRO		
FOOD CHOICES	CHO	PRO
1 Container GreekYogurt 1 Medium Banana ½ cup low fat granola ½ cup choc and ½ cup skim milk	75	22
1.25 cup vector + 1 cup fruit juice+ 1 cup skim milk	69	13
2 yogurt+1 pc fruit+ ½ bagel + 1 tsp jam	75	16
2 sl Banana Bread or 1 med low fat pumpkin muffin + 1 tbsp PB + ½ cup Kefir or ¾ cup greek yogurt	55	15
Elevate Bar + ½ cup juice	68	16

Hydration...Did you know?

Research conducted on elite Australian swimmers during **TAPER** found that male swimmers were losing approximately **415ml/h** and women were losing **365ml/h**.

Remember this is just in Taper!!



Monitor Your Urine Color!

AM I HYDRATED?

Urine Color Chart

1		
2		If your urine matches the colors 1, 2, or 3, you are properly hydrated.
3		Continue to consume fluids at the recommended amounts.
4		If your urine color is below the RED line, you are
5		<u>DEHYDRATED</u> and at risk for cramping and/or a heat illness!!
6		<u>YOU NEED TO DRINK MORE WATER!</u>
7		
8		

Staying Healthy!

- Calcium and Vit D
- Iron?
- B₁₂?
- Probiotics
- Omega 3
- Antioxidants!
 - Vit C, E





"Something from the supplement cart?"

76-91% of athletes use
dietary supplements
(Braun et al. 2009)



Sports Foods
Vitamin/minerals
Ergogenic Aids

Results from 3 Studies Conducted by the IOC (2000-02)

Table II. Results of the analysis of dietary supplements for anabolic agents carried out for the International Olympic Committee by the Cologne Doping Laboratory (Geyer et al., 2004).

Country	No. tested	No. "positive"	Percent "positive"
Netherlands	31	8	25.8
Austria	22	5	22.7
UK	37		
USA	240		
Italy	35		
Spain	29		
Germany	129		
Belgium	30		
France	30		
Norway	30		
Switzerland	13		
Sweden	6	0	0
Hungary	2	0	0
Total	634	94	14.8

Note: For each country where supplements were purchased, the table shows the number of samples tested, the number that contained prohibited steroids, and the fraction of the total this accounts for.

- 94 of 634 supplements (14.8%) contained prohormones not declared on labels "positive"

Bottom Line--Up to 1 in 5 supplements may be CONTAMINATED!

- 110 dietary supplements purchased, 14 contained undeclared caffeine, 2 with undeclared ephedrine

To Supplement or Not?

Issues to consider when deciding to use a sports food or supplement



Is it safe?



Is it legal?



Is there evidence
that it works??



Am I aware
of the correct
protocols of
how and when
to take it?



Can I afford it?

Always Remember..

- Moderate - High carbohydrate diet depending on training
- Balanced meals
- Variety of foods
- Fueling pre-workout and during
- Refueling post-workout
- Hydrate!



Further Individualized Follow Up

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